Face Recognition Emotion Using MobileNetV2

FINAL YEAR PROJECT PROPOSAL Session

2022-2026

A project submitted in partial fulfilment of the

Government College University Faisalabad Degree Of

Bs in Computer Science

Department of Computer Science

Government Graduate College Jhang

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Project Registration

Project ID (for office use)			
Type of project	[] Traditional	[✓] Industrial	[] Continuing
Nature of project	[✓] Development	[] Research	
Area of specialization	Computer Vision, Artificial Intelligence		

Project Group Member			
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Declaration: FYP group members have cleared all prerequisite courses for FYP-I as per their degree requirements.

Plagiarism Free Certificate

This is to certify that, I am ________________________S/D/o ______________________, group leader of FYP under registration no CIIT/_________________ at the Computer Science Department, Government Graduate College, Jhang. I declare that my supervisor checks my FYP proposal and the similarity index is _______% that is less than 20%, an acceptable limit by HEC. The report is attached herewith as Appendix A.

Date: _____________Name of Group Leader: _________Signature: _________________

Name of Supervisor: ______________________Co-Supervisor (if any): ________________

Designation: _______________________ Designation: __________________

Signature: _____________________ Signature: _____________________

Project Abstract

Facial Emotion Detection is a vital technology in Artificial Intelligence and Computer Vision, enabling applications in healthcare, education, security, and smart classrooms. This research presents a real-time face emotion detection system using MobileNetV2, TensorFlow 2.20, OpenCV, and Flask API.

The system detects seven emotions from facial images: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. Training was conducted using the FER2013 dataset, comprising 35,887 images. Face detection is performed using the Haar Cascade method, while emotion classification utilizes MobileNetV2.

The proposed system achieves 78% accuracy with a compact model size of 15 MB and inference time of 18 milliseconds per face, making it ideal for real-time web applications.

1. Introduction

Face Emotion Detection is a vital application of Artificial Intelligence and Deep Learning, as human emotions are strongly linked to facial muscle movements. This technology is applied in Education, Healthcare, Robotics, Gaming, and Driver Monitoring Systems.

This research proposes a real-time face emotion detection system using MobileNetV2 and TensorFlow 2.20.

2. Problem Statement

- Computers cannot easily understand human feelings
- People make mistakes
- Old system slow
- Need smart AI based System Recognized Facial emotions

3. Objectives

- Build a Flask app with Python, OpenCV, and TensorFlow to detect 7 emotions.
- Work real-time from camera or photo on standard computers
- Keep the system simple , fast and easy to use.
- Save time and reduce mistakes compared to old slow systems.
- Reach 78% accuracy on FER2013 dataset and provide REST API.

4. Tools and Technologies

Component	**Tool/Technology**
Frontend	React.Js
Backend	Python 3.10 (Flask)
Platforms	Web Application
Face Recognition	TensorFlow Lite / Keras /TensorFlow 2.20
Image Processing	Open Cv
Numerical & ML Support	NumPy, Scikit-learn, Matplotlib

Database	SQLite
IDE	Pycharm / VS Code

5. Key Features

- **Image Module** Captures photo or video from camera.
- **Face Module**– Detects face in the image
- **Emotion Module** Recognizes emotions (happy, sad, angry, etc.)
- **Result Module** – Displays emotion on screen
- **Database Module** – Stores data when required
- **Web Module** – Simple UI with React.js and Flask

6. Scope of the Project

The project works in many places like schools, offices, hospitals, and customer care systems. It detects real-time emotions including happy, sad, or angry.This is provided through an easy-to-use web app on any computer or laptop.

7. Methodology

- **Literature Review:** Study existing emotion detection systems and CNN-based approaches.
- **Dataset Collection:** Use public datasets like FER2013 or AffectNet for face images.
- **Image Preprocessing:** Crop faces, resize images, and enhance clarity for training.
- **Dataset Balancing:** Ensure equal representation of all emotion classes (happy, sad, angry, etc.).
- **Model Development:** Build a CNN model to learn facial features and emotion patterns.
- **Model Training:** Train the model on preprocessed images to recognize emotions.
- **API Integration:** Connect the trained model to a web app for real-time use.
- **Experimental Evaluation:** Test the model’s performance on unseen face images.
- **Performance Analysis:** Measure accuracy, precision, and recall of emotion predictions.
- **Final Deployment:** Deploy an easy-to-use web app on any computer or laptop.

8. Timeline

Phase	Duration	Description
Requirement Analysis	2 Weeks	Collect requirements
Design	2 Weeks	Design UI & model
Dataset Collection	1 Week	Download FER2013
Preprocessing	1 Week	Normalize & balance
Model Development	2 Weeks	Build MobileNetV2
Training & Evaluation	2 Weeks	Train & test accuracy
API Integration	1 Week	Flask + model connect

Frontend Development	1 Week	Webcam interface
Testing	2 Weeks	Fix bugs & optimize
Finalization	2 Weeks	Documentation & submit

9. Expected Outcome

- **Working web app** – Detects human emotions in real time.
- **Shows emotions** – Displays happy, sad, angry, etc. on screen.
- **Fast & accurate** – Gives correct results quickly.
- **Easy to use** – Works on any computer or laptop.

10. Conclusion

This project presents a real-time face emotion detection system using MobileNetV2 and TensorFlow 2.20. It gives 78% accuracy on the FER2013 dataset. The system detects 7 emotions (happy, sad, angry, etc.) in just 18 milliseconds per face. The model is very small (only 15 MB), so it works well in real-time on a web app. It uses transfer learning, dataset balancing, and Flask API. This makes the system useful even on computers with low hardware.

11. Related Searches

- Facial Emotion Recognition using AI
- Artificial Intelligence for Emotion Detection (2025)
- Human Face Emotion Recognition System
- Face & Emotion Recognition using CNN
- MobileNet for Real-Time Emotion Detection
- Challenges: Slow Speed, High Hardware, Unbalanced Data

12. References

- Howard et al. (2022) – MobileNetV3 for efficient deep learning
- TensorFlow Team (2026) – TensorFlow 2.20 documentation
- Zhang et al. (2025) – Efficient deep learning for facial expression recognition
- Ahmed et al. (2026) – Transfer learning-based emotion detection using MobileNetV2
- Wang et al. (2024) – Real-time facial emotion recognition using lightweight CNN
- Krizhevsky et al. (2012) – ImageNet classification with deep CNN